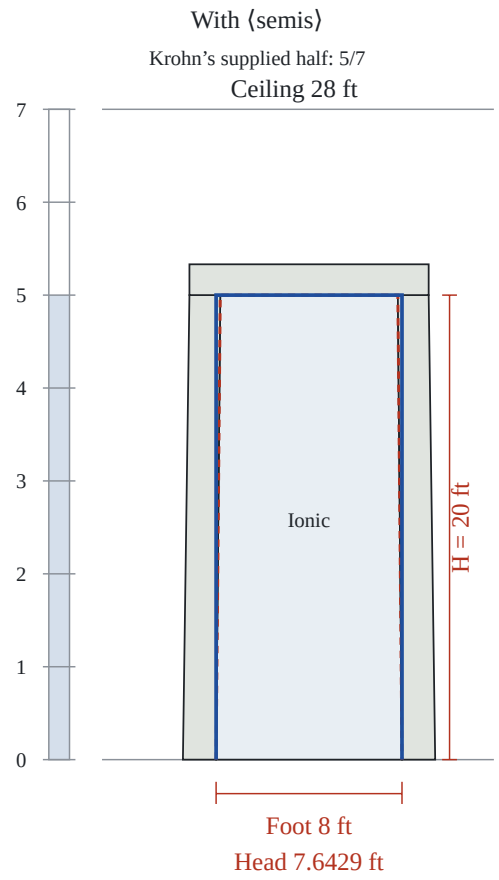
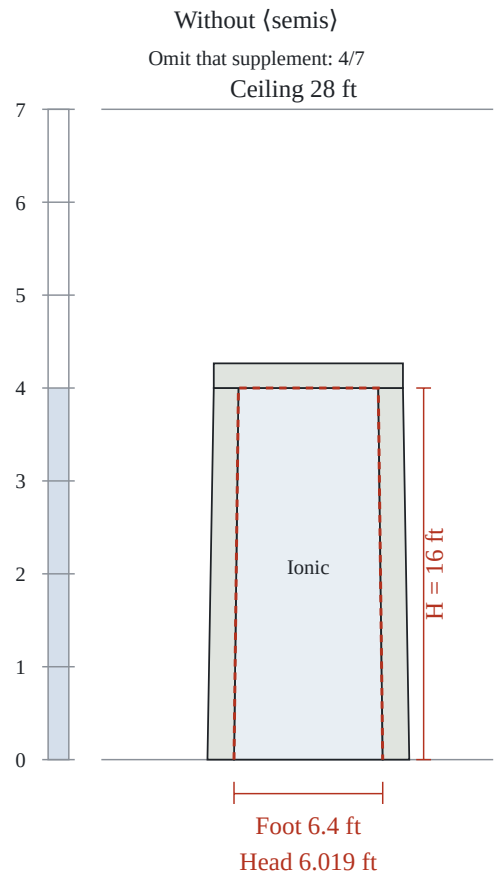
A half changes the doorway

Vitruvius's doorway proportions, read with and without the supplied half. Ceiling 28 Roman feet; Ionic; selected reading with ⟨semis⟩ (5/7). Selected opening height 20 ft. Over 16 through 25 ft: one quarter of a jamb.

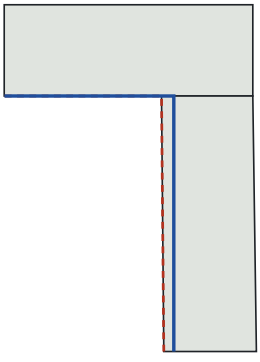


Selected reading. Solid blue: your trial. Dashed red: computed rule.



The other height reading, with its own contraction band.

Upper right corner, enlarged



Both axes enlarged equally. Stone follows the rule.
Trial differs by 0.1786 ft at this corner.

Measurements for the selected reading

Measure	Roman feet	Basis
Ceiling height	28	chosen
Opening height	20	5/7 of ceiling; supplied semis
Opening width at foot	8	2/5 of opening height
Jamb width at foot	1.4286	opening height / 14; IV.6.3
Jamb width at head	1.3265	13/14 of lower jamb width; IV.6.2
Whole opening contraction	0.3571	lower jamb width / 4; applied to total width
Opening width at head	7.6429	lower width minus whole contraction
Each upper corner inset	0.1786	half the whole contraction
Lintel height	1.3265	upper jamb width; IV.6.2
Trial upper width	8	learner trial; not an ancient prescription
Trial upper width minus rule	0.3571	positive means too wide

Choices and limits

- A reading of Vitruvius IV.6.1–3, not a reconstruction of a particular ancient doorway.
- The height comparison omits only Krohn's second, supplied semis. Other editorial choices stay fixed.
- The opening contraction is subtracted once from the whole width, half from each side; the lower jamb supplies the unit.
- The shared endpoints 16, 25 and 30 feet belong to the lower contraction band on this desk. The text does not uniquely settle these endpoints.
- The Ionic jamb is H/14, as printed on Krohn p. 89. The Doric jamb uses that same value as a stated drawing assumption across Krohn's lacuna.
- Jamb widths are measured horizontally on the face. The upper jamb retains 13/14 of the lower width. The lintel height equals the upper jamb width.
- The casing follows the computed rule. Solid blue is the learner's trial, which can cross into that casing. Both main drawings share a scale; the corner enlargement uses the same scale on both axes.
- Mouldings, frieze, consoles, door leaves, wall depth, optical perception and structural performance are not modelled. Roman feet are used without a modern metric conversion.

Words in the drawing

lvmen · The opening. luminis: ‘of the opening’. The clear space through the wall. The Doric width sentence measures this empty space at its foot; adding two jambs would measure something else. The more general Latin sense is ‘light’. (IV.6.1, p. 88)

antepagmentvm · The casing. antepagmenti: ‘of the casing’. The doorway surround; here the side pieces are the jambs. Its width supplies the unit for narrowing the opening. Its own narrowing follows a different fraction, one fourteenth. (IV.6.1–3, pp. 88–89)

svpercilivm · The lintel. supercilii: ‘of the lintel’. An ‘eyebrow’ above the doorway. The text relates its thickness to the upper end of the casing. The drawing shows that relation but leaves the mouldings and the frieze above it aside. (IV.6.2, p. 88)

lacvnaria · The coffered ceiling. ad lacunaria: ‘to the coffered ceiling’. This is the upper end of the height being divided. The opening height is a fraction of the distance from the pavement to this ceiling; it is not a fraction of the columns or the roof ridge. (IV.6.1, p. 88)

contrahatvr · Let it narrow. Present passive subjunctive of contraho. A prescription: ‘let it be drawn together’. Identify what is being narrowed and what supplies the fraction. The opening narrows by a fraction of a jamb; the jamb narrows by a fraction of itself. (IV.6.1–2, p. 88)

valvae · The door leaves. valvarum: ‘of the doors’. The doors that close the opening. The Latin distinguishes these from the lumen. Door-leaf construction appears later in IV.6; the drawing deliberately leaves the opening empty. (IV.6.1, p. 88)

θύρωμα · A Greek word inside the Latin. Greek θυρωμάτων, genitive plural of θύρωμα. Vitruvius names Doric, Ionic and Attic kinds of doorway with this Greek genitive plural, ‘of doorways’. Krohn prints the word in Latin letters on p. 88; the accented Greek here is the dictionary form, not a quotation of Greek lettering on the scan. (IV.6.1, p. 88)

Read the instructions

IV.6.1 · Three doorway kinds

genera svnt enim thyromaton haec: doricvm, ionicvm, atticvrges.

For these are the kinds of doorway: Doric, Ionic and Attic.

Thyromaton is Greek in Latin letters. This desk draws only the Doric and Ionic proportions; it does not supply an Attic doorway.

Krohn 1912, printed p. 88.

IV.6.1 · A supplied half

lvmen avtem thyretri constitvatvr sic, vti qvae altitvdo aedis a pavimento ad lacvnaria fverit, dividatvr in partes tres semis et ex eis dvae partes ⟨semis⟩ lvmini valvarvm altitvdine constitvantvr.

Set the opening of the doorway as follows: divide the temple's height from the pavement to the coffered ceiling into three and a half parts; of these, let two ⟨and a half⟩ parts establish the height of the opening for the doors.

Krohn prints the second semis inside angle brackets. His apparatus credits its addition to Rode. With it, (2½)/(3½) = 5/7; omit only that supplement and 2/(3½) = 4/7. These are two readings of this instruction, not two complete recovered buildings or two uniform manuscript traditions.

Krohn 1912, printed p. 88.

IV.6.1 · The lower opening

haec avtem dividatvr in partes xii et ex eis qvinqve et dimidia latitvdo lvminis fiat in imo.

Divide this [height] into twelve parts, and let five and a half of them be the width of the opening at the bottom.

The Doric lower opening is 5½/12 = 11/24 of its height. Luminis means 'of the opening': this width excludes both stone jambs.

Krohn 1912, printed p. 88.

IV.6.1 · The opening's contraction

et in svmmo contrahatvr, si erit lvmen ab imo ad sedecim pedes, antepagmenti iii parte; ⟨si⟩ xvi pedvm ad xxv, svperior pars lvminis contrahatvr antepagmenti parte iiii; si ab pedibvs xxv ad xxx, svmma pars contrahatvr antepagmenti parte viii. reliqva, qvo altiora ervnt, ad perpendicvlvm videntvr oportere conlocari.

At the top let it narrow, if the opening is up to sixteen feet high from the bottom, by a third of the casing; from sixteen feet to twenty-five, let the upper part of the opening narrow by a quarter of the casing; from twenty-five to thirty feet, by an eighth. The remaining, taller ones seem to require perpendicular sides.

The fraction is of the jamb, not the opening or its height. This desk subtracts it once from the whole opening width and splits that loss equally between the two sides. The ranges share endpoints: this desk keeps 16, 25 and 30 in the lower band. That boundary convention and the symmetric allocation are modelling decisions. No interpolation between bands is implied.

Krohn 1912, printed p. 88.

IV.6.2 · The casing's own taper

ipsa avtem antepagmenta * * * * contrahantvr in svmmo svae crassitvdinis xiiii parte. svpercilii crassitvdo, qvanta antepagmentorvm in svmma parte erit crassitvdo.

The casings themselves [gap in the text] are to narrow at the top by a fourteenth of their own width. The lintel's thickness is to be as much as the casings' width at their upper end.

The stars reproduce Krohn's marked lacuna, not a passage omitted for brevity. The jamb loses 1/14 of its own lower width, independently of the opening's contraction. The desk draws the lintel height as that upper jamb width and omits the overdoor mouldings.

Krohn 1912, printed p. 88.

IV.6.3 · The Ionic opening and casing

Sin avtem ionico genere fvtvra ervnt, lvmen altvm ad evndem modvm qvemadmodvm in doricis fieri videtvr. latitvdo constitvatvr, vt altitvdo dividatvr in partes dvas et dimidiam, eivsqve partis vnivs ima lvminis fiat latitvdo. contractvrae ita vti in doricis. crassitvdo antepagmentorvm ⟨ex⟩ altitvdine lvminis in fronte xiiii parte

If they are to be Ionic, the opening's height seems to be made in the same way as in Doric. Set the width by dividing the height into two and a half parts and giving the lower opening one of them. The contractions are as in Doric. The width of the casings on their face is a fourteenth of the opening's height.

The print on p. 89 reads XIIII, fourteen, rather than XII, twelve. This fixes the Ionic lower jamb at H/14. For Doric, the desk explicitly borrows that Ionic value across the lacuna; it does not silently supply missing Latin. Ionic lower width is H/2½ = 2H/5. Consoles, fasciae and door leaves are outside this diagram.

Krohn 1912, printed p. 89.

Keep working with the drawing

This file includes both diagrams, the corner, measurements, Latin, new English translations, complete source pages, four font subsets and licenses. It uses no scripts and opens offline. External links need a connection.

Reopen these choices in Embater.

Vitruvius, *De architectura*, ed. F. Krohn (Teubner, 1912), complete printed p. 88, PDF page 106. Public domain. Original scan. SHA-256 bd7e110dbd0db15f159cd48cde18910f9da0fb83128d58ae5beac673cbc115dd.

VI Ostiorum autem et eorum antepagmentorum in aedibus hae sunt rationes, uti primum constituatur, quot genera sint [futurae]. genera sunt enim thyromaton haec: doricum, ionicum, atticurges.

Horum ⟨dorici generis⟩ symmetriae conspiciuntur his rationibus, uti corona summa, quae supra antepagmentum superius inponitur, aeque librata sit capitulis summis columnarum, quae in pronao fuerint. lumen autem thyretri constituatur sic, uti quae altitudo aedis a pavimento ad lacunaria fuerit, dividatur in partes tres semis et ex eis duae partes ⟨semis⟩ lumini valvarum altitudine constituantur. haec autem dividatur in partes XII et ex eis 97 quinque et dimidia latitudo luminis fiat in imo. et in summo contrahatur, si erit lumen ab imo ad sedecim pedes, antepagmenti III parte; ⟨si⟩ XVI pedum ad XXV, superior pars luminis contrahatur antepagmenti parte IIII; si ab pedibus XXV ad XXX, summa pars contrahatur antepagmenti parte VIII. reliqua, quo altiora erunt, ad 2 perpendiculum videntur oportere conlocari. ipsa autem antepagmenta * * * * contrahantur in summo suae crassitudinis XIIII parte. supercilii crassitudo, quanta antepagmentorum in summa parte erit crassitudo. cymatium faciundum est antepagmenti parte sexta; proiectura autem, quanta est eius crassitudo. sculpendum est cymatium lesbium cum astragalo. supra cymatium, quod erit in supercilio, conlocandum est hyperthyrum crassitudine supercilii, et in eo scalpendum est cymatium doricum, astragalum lesbium sima scalptura. corona plana cum cymatio; proiectura autem eius erit quanta altitudo. super-

---

2 constituatur, quot genera sint *Kr*: constituantur quod (quo *GS*) genere sint futurae *x* (*de delendo* futurae *cf. pag. 253,26 et 266,2*) 4 attigurges *x* 5 *add. Joc* 7 inponetur *HS* 8 thyretri *Ro*: hypetri *x* 10 III semis *G* 11 partis *H add. Rode* lumini *Joc*: lumine *x* 13 latitudo H^cSG^c: altitudo *a. c. HG* 15 pedis *H* si *add. Schn* 16 quarta parte *G* 17 ad pedibus *H et a. ras. S*, a pedibus GS^c 23 faciendum *GS* 28 astragulū lespiū *S*

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cilii, quod supra antepagmenta inponitur, dextra atque sinistra proiecturae sic sunt faciundae, uti crepidines ex-
(20) currant et in unguem ipso cymatio coniungantur.

Sin autem ionico genere futura erunt, lumen altum ad 3
5 eundem modum quemadmodum in doricis fieri videtur. latitudo constituatur, ut altitudo dividatur in partes duas et dimidiam, eiusque partis unius ima luminis fiat latitudo.
(25) contracturae ita uti in doricis. crassitudo antepagmentorum ⟨ex⟩ altitudine luminis in fronte XIIII parte, cy-
98 matium huius crassitudinis sexta. reliqua pars praeter cy-
11 matium dividatur in partes XII. harum trium prima corsa fiat cum astragalo, secunda quattuor, tertia quinque, et eae aeque corsae cum astragalis circumcurrant. hyper- 4
(5) thyra autem ad eundem modum componantur quemad-
15 modum in doricis. pro crepidinibus ancones, sive parotides vocantur, excalptae dextra ac sinistra praependeant ad imi supercilii libramentum, praeter folium. eae habeant in fronte crassitudinem ex antepagmenti tribus partibus,
(10) imo quarta parte graciliore quam superiora.

Fores ita compingantur, uti scapi cardinales sint ex
20 latitudine luminis totius XII parte. inter duos scapos tympana ex XII partibus habeant ternas partes. inpagibus 5 distributiones ita fient, uti divisis altitudinibus in partes V
(15) duae superiori, tres inferiori designentur. super medium
25 medii inpages conlocentur, ex reliquis alii in summo, alii in imo compingantur. altitudo inpagis fiat tympani tertia parte, cymatium sexta parte inpagis. scaporum latitudines inpagis dimidia parte, item replum de inpage dimidia et

1 ac sin. *G* 2 faciendae *G* 3 unguem *Schn*: ungue *x* 7 ima luminis *G et in ras. S*: simalum in his *H* 8 ctraiecturę *S* 9 ex *add. Kr* (*cf. 98, 8 et 11*) 11 dividatur *ed*: -ditur *x* harum *Joc*: horum *x* 12 et eae *Ro*: ex ea *x* 13 hyperthyra *Joc*: hypetra *HS*, hipetre *G* (-ę *G*c) 15 pro crepidinibus *Kr*: protis pedibus *x* parotides *Schn*: protides *HG*, ptides *S* 16 excalptae *Ro*: excalpta *H*, exscalpta *S*, etscalpta *G* 18 antepagmenti *Joc*: -tis *x* 19 imo *Kr*: inimo (*cum* in *add.*) *x* 21 altitudine *HS* 22 duodecī *S* inpagibus *ed*: inpaginibus *x* 24 III *G* 25 ex *ed*: et (&) *x* 27 scaphorum *HS*

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